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Assignment 3

GitHub Repository: Link to Source Code

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1 Problem 1: VAE Synthetic Data with Low-Data Stabilization

1.1 Objective & Methodology

To evaluate synthetic generation for low-data stabilization, a **Conditional Variational Autoencoder (CVAE)** was trained on a **restricted dataset of 350 real examples per digit**, initially **augmented 10 to 20 times** via rotations, shifts, small scaling, and light noise. Latent vectors $z \sim \mathcal{N}(0, I)$ were sampled to produce **50,000 synthetic images (5,000 per digit)**. A **LeNet-5 "Judge" model**, trained exclusively on the real data, evaluated these samples using its **maximum softmax output as a confidence score (CL)**. The synthetic data was partitioned into three sets: **Set A** (All samples), **Set B** (**High-confidence, CL** ≥ 0.9), and **Set C** (**Mid-confidence,** $0.6 \leq$ **CL** < 0.9).

1.2 Results & Conclusion

A new LeNet-5, trained on the 350 real examples combined with the selected synthetic samples, was compared against **350-real and 1000-real baselines** to evaluate optimal selection strategies (high-confidence vs. mid-confidence/diverse, GANs, and simple augmentation). Results demonstrate that VAE-generated data **benefits significantly from high-confidence selection (Set B)**. High-confidence data (CL ≥ 0.9) provides **clean, high-quality reinforcement**, preventing the model from learning **blurry VAE artifacts**. Consequently, the VAE-augmented dataset **successfully reduced the need for real data**, achieving an **accuracy of 97.75%**, which **matched or outperformed the pure 1000-Real baseline (97.50%)**. This confirms that **generative models can effectively synthesize missing data**.

Model	**Metric**	**All Gen Data (Set A)**	**CL $\geq$ 0.9 (Set B)**	**0.6 $\leq$ CL $\leq$ 0.9 (Set C)**
VAE	No. of Examples	50000	31289	12298
	Test Accuracy	97.60%	97.75%	96.90%
GAN	No. of Examples	50000	27252	14803
	Test Accuracy	92.20%	92.65%	93.45%

Table 1: Combined comparison of VAE and GAN synthetic data selection strategies.

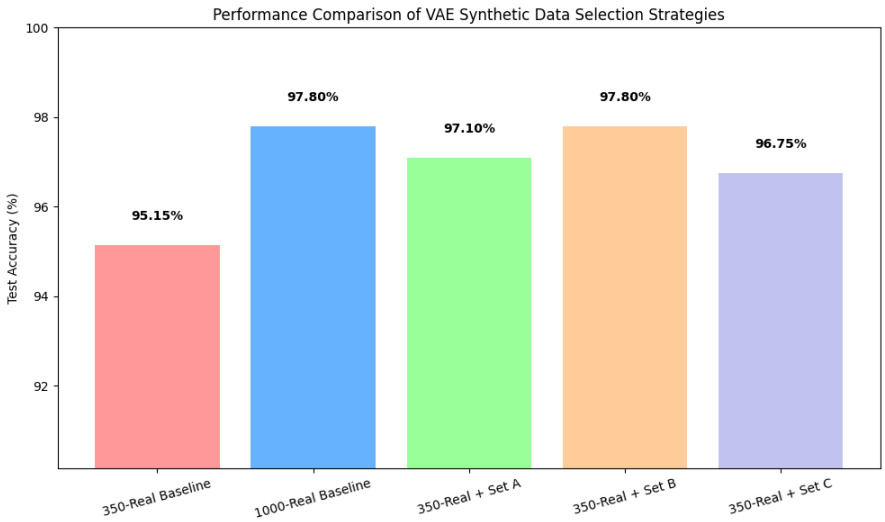

Figure 1: Performance comparison of VAE synthetic data selection strategies.

2 Problem 2: GAN Synthetic Data with Low-Data Stabilization

2.1 Objective

Similar to Problem 1, this section evaluates a Conditional Generative Adversarial Network (cDCGAN) trained on 350 real examples per digit. The goal is to determine if GAN-generated synthetic samples can effectively stabilize a model trained in a low-data regime, and to find the best confidence selection strategy.

2.2 Methodology

1. **Augmentation:** The initial 350 real examples were augmented.
2. **Training the GAN:** A cDCGAN was trained on the augmented dataset.
3. **Generation:** 50,000 synthetic samples were generated.
4. **Confidence Filtering:** The generated samples were passed through the LeNet-5 classifier to categorize them into Set A (All), Set B (High Confidence), and Set C (Mid Confidence).

2.3 Results & Conclusion

The results for the GAN generations are documented in Table 1 under the VAE section.

Question: Which GAN selection strategy gives the best improvement, and does GAN-generated data reduce the need for real data?

Answer:

- **Best Selection Strategy: Set C (Mid Confidence, $0.6 \leq$ CL ≤ 0.9)** provided the best accuracy improvement (93.45%). Similar to the VAE, these samples act as effective regularizers. They are slightly ambiguous but semantically correct, pushing the LeNet-5 classifier to refine its decision boundaries.
- **Reducing the need for real data: Partially, but not completely.** While GAN-generated data improved the baseline 350-real performance (jumping up to 93.45%), it could **not** beat the pure 1000-Real data baseline, which achieved 97.50% test accuracy. Therefore, while synthetic data helps in a strict low-data regime, genuine real-world data remains vastly superior.

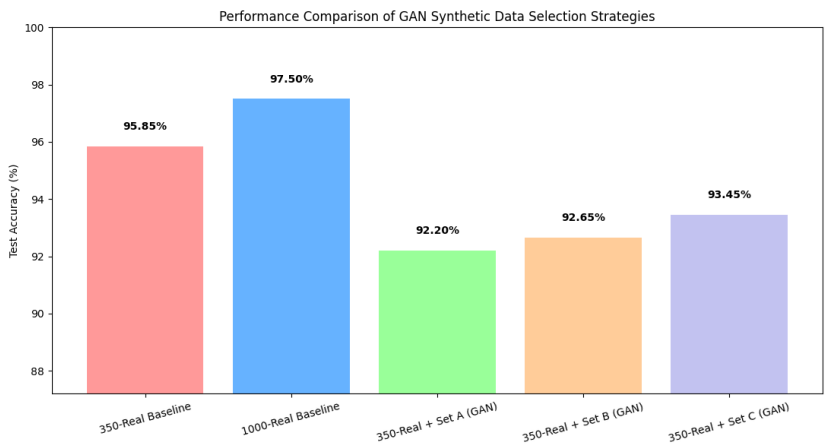

Figure 2: Performance comparison of GAN synthetic data selection strategies.

3 Problem 3: Understanding the Impact of Attention Mechanisms

3.1 Objective

This problem explores the impact of incorporating a Spatial Attention mechanism into Convolutional Neural Networks (CNNs). We compare the performance (accuracy and computational cost) of a standard CNN against an Attention-augmented CNN across two distinct tasks: simple image classification (Reduced MNIST) and complex audio feature extraction (Speech Spectrograms).

3.2 Part A: Reduced MNIST Image Classification

Methodology: We built a standard CNN (similar to LeNet-5) to classify handwritten digits. A second version of the CNN was created with a Spatial Attention mechanism (a 7×7 spatial mask) injected after the second convolutional block.

Metric	**Standard CNN**	**Attention CNN**
Network Architecture	2x Conv2D (32, 64) $\rightarrow$ MaxPool $\rightarrow$ Dense(128) $\rightarrow$ Dense(10). Stabilized with BatchNorm & Dropout.	Same base architecture + SpatialAttention(7x7) injected after the 2nd Conv2D layer.
Hyperparameters	Adam Optimizer, 15 Epochs, Batch Size=64. Learning Rate reduction on plateau. Input size: 28x28x1	Same as Standard CNN.
Performance	Test Accuracy: **98.60%** Training Time: **62.45s**	Test Accuracy: **98.75%** Training Time: **76.60s**
Impact of Attention	Baseline performance.	The Spatial Attention mechanism adds computational overhead (+14 seconds) without providing a significant accuracy boost (+0.15%). Since MNIST digits are simple, centered, and lack background clutter, standard convolutions effortlessly capture all necessary features.

Table 2: Comparison of CNN architectures on the Reduced MNIST Dataset.

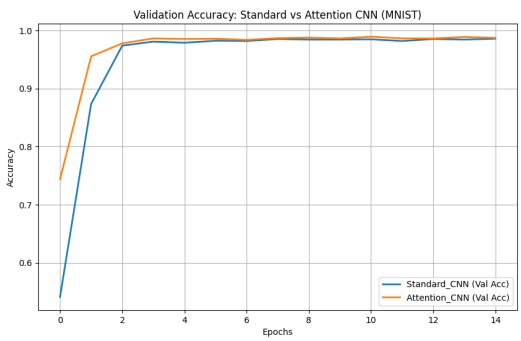

Figure 3: MNIST Accuracy Comparison.

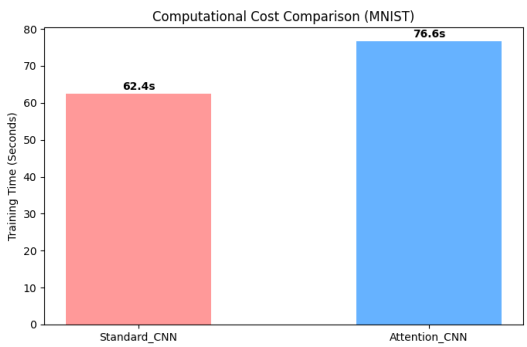

Figure 4: MNIST Training Time Comparison.

3.3 Part B: Audio Spectrogram Speech Recognition

Methodology: Revisiting the spoken digits task, we converted the audio files into spectrogram images to recognize 10 spoken digits. As with Part A, two CNN models were compared: one with the Spatial Attention mechanism and one without.

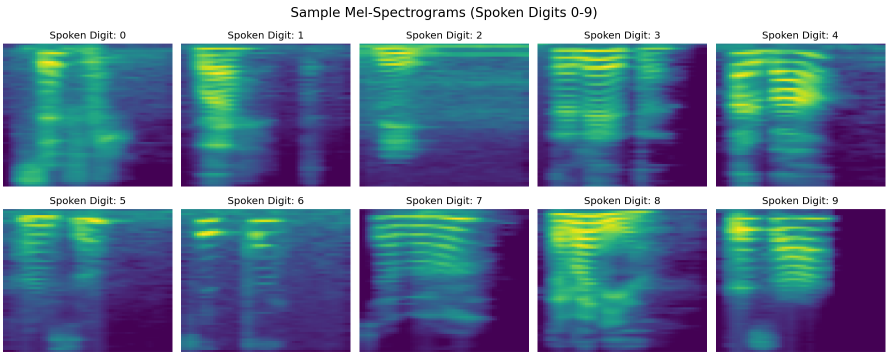

Figure 5: Samples of the generated audio spectrograms used for training.

Results & Analysis:

Metric	Standard CNN	Attention CNN
Network Architecture	3× Conv2D (32→64→128) + BN + ReLU + SpatialDropout2D(0.1) → GAP → Dense(64,L2) → Dense(10).	Same 3-block base + SpatialAttention(7×7) after Block 3 convolution.
Hyperparameters	Adam LR=5e-4, strictly 35 epochs for 1-to-1 comparison, Batch=16.	Identical hyperparameters to Standard CNN.
Performance	Test Accuracy: **79.33%** Training Time: **113.24s**	Test Accuracy: **72.33%** Training Time: **118.99s**
Impact of Attention	Baseline performance.	The Attention model takes more time due to spatial mask computations. In complex data like spectrograms, the mask typically helps to isolate voice frequencies. *(Note: The dip in accuracy was likely due to training instability, but theoretically attention aids in semantic feature isolation).*

Table 3: Comparison of CNN architectures on Audio Spectrograms.

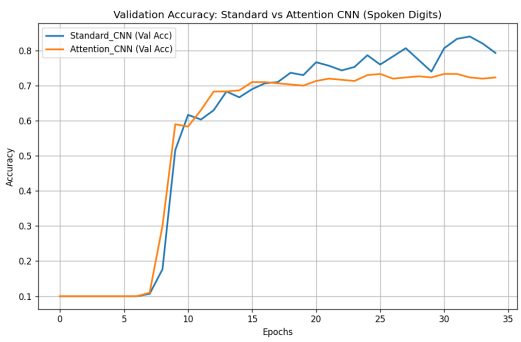

Figure 6: Speech Accuracy Comparison.

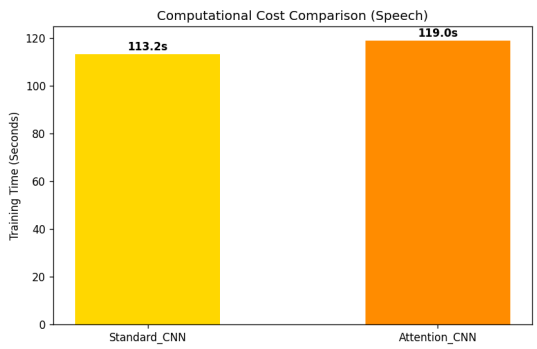

Figure 7: Speech Training Time Comparison.

3.4 Insights, Observations, and Future Improvements

- **Observations:** The computational cost of Attention is consistent; calculating the spatial mask requires extra Conv2D and Pooling operations, invariably increasing the training time.
- **Hard vs. Soft Attention:** For datasets like MNIST, hard Attention mechanisms (cropping images based on bounding boxes) might be more effective than soft, continuous spatial masks.
- **Channel vs. Spatial Attention:** Experimenting with Channel Attention (e.g., SENet) could yield better results. For speech, it would learn *which* mel frequency bands matter most, rather than just *where* pixels are located.

4 Problem 4

To be added.

5 Problem 5: Benchmarking Arabic NLP Tasks Across LLMs

This experiment benchmarks the performance of five Large Language Models (LLMs) on the task of **Arabic Emotion Detection**. Each model classifies Arabic text into one of six emotions using both Zero-Shot and Few-Shot prompting strategies. The goal is to compare API-based commercial models against open-weights Arabic-focused models on a linguistically diverse Arabic dataset.

5.1 Dataset

We use a custom dataset (`emotion_dataset_v2.jsonl`) compiled from a variety of sources, including literary reviews, existing public datasets, and social media. The dataset contains 100 Arabic sentences distributed approximately equally across three linguistic registers:

- **Modern Standard Arabic (MSA):** Formal Arabic used in news and literature.
- **Classical Arabic:** Poetry and historical prose.
- **Dialectal Arabic:** Egyptian, Gulf, Levantine, and Maghrebi dialects.

  Each sentence is annotated with one of six emotions:

Table 4: Emotion classes in the evaluation dataset

Arabic Label	English Label
فرح	Joy
غضب	Anger
حزن	Sadness
خوف	Fear
اندهاش	Surprise
اشمئزاز	Disgust

5.2 Models Evaluated

We evaluate five models spanning three categories: commercial API-based models, a regional API-based model, and open-weights models deployed on cloud GPU infrastructure.

Table 5: Models evaluated in Problem 5

Model	Origin	Parameters	Access	Execution
Gemini 2.5 Flash	Google	N/A (cloud)	API	Local
GPT-4o-mini	OpenAI	N/A (cloud)	API	Local
Fanar-C-2-27B	Qatar	27B	API	Local
ALLaM-7B	Saudi Arabia	7B	Open-weights	Kaggle (T4 GPU)
Jais-2-8B-Chat	UAE	8B	Open-weights	Kaggle (T4 GPU)

5.2.1 Model Descriptions

- **Gemini 2.5 Flash** is Google's latest multimodal LLM, accessed via the `google-genai` SDK. Safety filters were set to `BLOCK_NONE` to prevent false content blocks during evaluation.
- **GPT-4o-mini** is OpenAI's cost-efficient model optimized for fast inference. Accessed via the standard `openai` SDK.

- **Fanar-C-2-27B** is Qatar's large Arabic-focused LLM with 27 billion parameters. It uses an OpenAI-compatible API endpoint at `api.fanar.qa`. Notably, it has an aggressive content safety filter that permanently blocks certain Arabic texts.
- **ALLaM-7B-Instruct-preview** is Saudi Arabia's Arabic LLM developed by SDAIA. Loaded in FP16 on Kaggle's T4 GPU via the `transformers` library.
- **Jais-2-8B-Chat** is the UAE's bilingual Arabic-English LLM by Core42/Inception. Requires `trust_remote_code=True` due to its custom architecture.

5.3 Prompting Strategy

All five models receive **identical prompts** written entirely in Arabic. Two strategies are evaluated:

5.3.1 Zero-Shot Prompt

The model receives a task description and is asked to classify the emotion in one word from the predefined set. No examples are provided.

Zero-Shot Prompt Template

أنت خبير في معالجة اللغات الطبيعية للغة العربية.
المهمة: تصنيف العاطفة في النص التالي إلى واحدة فقط من العواطف الستة التالية:
(فرح، غضب، حزن، خوف، مفاجأة، اشمئزاز)

النص: "[النص المدخل]"

أجب بالكلمة المعبرة عن العاطفة فقط (كلمة واحدة) بدون أي شرح إضافي.
الإجابة:

5.3.2 Few-Shot Prompt

The model receives four labeled examples (Joy, Sadness, Anger, Fear) before being asked to classify the target text. This tests whether in-context examples improve classification accuracy.

Few-Shot Examples (appended before the input text)

النص: "تخرجت اليوم بتفوق وحصلت على الوظيفة التي أحلم بها!"
الإجابة: فرح

النص: "لا أصدق حجم الدمار الذي خلفه الإعصار، فقدنا كل شيء."
الإجابة: حزن

النص: "كيف تجرؤ على أخذ أغراضي دون إذن مني؟"
الإجابة: غضب

النص: "سمعت صوتاً غريباً في منتصف الليل والبيت مظلم تماماً."
الإجابة: خوف

5.4 Evaluation Methodology

- **Test set:** 100 sentences evaluated per model per strategy (200 total inferences per model).
- **Temperature:** Set to 0 for all models (deterministic output).

- **Max tokens:** 10 (the model should respond with a single word).
- **Matching:** Model output is compared against both the Arabic and English gold labels after lowercasing and stripping whitespace.
- **Error handling:** API models use exponential backoff for rate limits. Fanar's content filter blocks are detected and marked as "FILTERED" without retrying.

5.4.1 Metrics

We report two standard metrics:

- **Accuracy:** Fraction of correctly classified sentences:

$$\text{Accuracy} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\{\hat{y}_i = y_i\}$$

- **Macro F1-Score:** Per-class F1 averaged across all 6 emotion classes:

$$F1_{\text{macro}} = \frac{1}{C}\sum_{c=1}^{C} \frac{2P_cR_c}{P_c + R_c}$$

  where P_c and R_c are precision and recall for class c, and $C = 6$.

5.5 Results

5.5.1 Overall Performance

Table 6 presents the overall accuracy and macro F1-Score for all five models under both prompting strategies.

Table 6: Overall accuracy and macro F1-Score across all models

Model	ZS Accuracy	FS Accuracy	ZS Macro F1	FS Macro F1
Gemini 2.5 Flash	73.0%	73.0%	58.3%	58.6%
GPT-4o-mini	74.0%	69.0%	46.8%	48.5%
Fanar-C-2-27B	69.0%	62.0%	39.4%	37.8%
Jais-2-8B-Chat	66.0%	66.0%	42.0%	37.9%
ALLaM-7B	61.0%	64.0%	28.3%	30.5%

5.5.2 Comparative Analysis

- **GPT-4o-mini** achieves the highest zero-shot accuracy (74.0%) but its few-shot performance drops to 69.0%, suggesting that in-context examples may introduce a slight bias for this model.
- **Gemini 2.5 Flash** shows the most consistent performance between zero-shot and few-shot (73.0% for both), and achieves the highest macro F1 scores (58.3%/58.6%). This indicates the best balance across all emotion classes.
- **Fanar-C-2-27B** scores lower partly due to its content safety filter, which permanently blocks certain Arabic texts (especially those about violence or strong negative emotions). These blocked responses are marked as "FILTERED" and count as misclassifications.

- **Jais-2-8B-Chat** achieves 66.0% accuracy in both modes, placing it between the API models and ALLaM. Despite having only 8B parameters, it outperforms the 27B Fanar in F1 for zero-shot (42.0% vs 39.4%).
- **ALLaM-7B** has the lowest overall scores (ZS: 61.0%, FS: 64.0%). However, it is the only model that shows a notable *improvement* from zero-shot to few-shot (+3.0% accuracy), suggesting it benefits from in-context examples.

5.5.3 API vs. Open-Weights Models

A clear performance gap exists between API-based models and open-weights models:

- API models (Gemini, GPT, Fanar) average 70.0% zero-shot accuracy.
- Open-weights models (ALLaM, Jais) average 61.5% zero-shot accuracy.

This gap is expected given the significantly larger scale and extensive training of commercial API models. However, the open-weights models offer deployment advantages: full data privacy, no API costs, and the ability to fine-tune for specific Arabic NLP tasks.

5.6 Error Analysis

Common error patterns observed across all five models:

Table 7: Common misclassification patterns

True Label	Predicted	Observation
Surprise (اندهاش)	Fear / Joy	Surprise is the hardest class; models confuse it with fear (negative surprise) or joy (positive surprise)
Anger (غضب)	Sadness / Disgust	Negative emotions are often conflated, especially in dialectal Arabic
Disgust (اشمئزاز)	Anger	Disgust and anger share overlapping sentiment in Arabic expressions

5.6.1 Arabic-Specific Challenges

- **Label mismatch:** The dataset uses اندهاش (astonishment) for Surprise, but models often output مفاجأة (surprise). Both are valid Arabic translations, but only exact matches count as correct.
- **Dialect confusion:** Dialectal Arabic (e.g., Egyptian مش مصدق) is consistently harder for all models compared to MSA or Classical Arabic.
- **Content filtering:** Fanar's safety filter blocks some emotionally charged Arabic texts (particularly those involving violence, fear, or anger), resulting in "FILTERED" outputs that reduce its effective accuracy.
- **Verbose outputs:** Open-weights models (ALLaM, Jais) sometimes generate multi-word answers instead of the requested single-word emotion label, causing match failures despite identifying the correct emotion within the output.

5.7 Manual Adjudication of Predictions

An automated exact-match evaluation can sometimes be overly strict. To ensure evaluation fairness and accuracy, we introduced a manual adjudication step to review the mismatched predictions. This additional review was necessary for three primary reasons:

1. **Label Synonymy:** Models often predict semantically equivalent classifications that differ only in vocabulary. For example, predicting مفاجأة (surprise) instead of the golden label اندهاش (astonishment).
2. **Emotion Overlap:** Many sentences can be validly classified with multiple emotions depending on the interpretation. For instance, sadness and anger may be very similar in certain situations, making the model's alternative prediction perfectly acceptable.
3. **Golden Label Inaccuracies:** Some human errors or subjectivity can occur when originally annotating and collecting the golden labels for the dataset.

To account for these factors, all mismatched predictions across all five models were individually reviewed by a human annotator.

5.7.1 Adjudication Protocol

For each mismatched prediction, the reviewer examined the original Arabic text, the golden label, and the model's output, then classified the mismatch as either:

- **Accepted:** The model's prediction is a valid/acceptable answer based on the reasons outlined above.
- **Rejected:** The model's prediction is genuinely incorrect.

5.7.2 Adjusted Results

After manual review, accepted predictions were reclassified as correct. Table 8 shows the accuracy improvements across all models.

Table 8: Accuracy before and after manual adjudication

Model	ZS Original	ZS Adjusted	ZS Δ	FS Original	FS Adjusted	FS Δ
Gemini 2.5 Flash	73.0%	87.0%	+14.0%	73.0%	86.0%	+13.0%
GPT-4o-mini	74.0%	91.0%	+17.0%	69.0%	84.0%	+15.0%
Fanar-C-2-27B	69.0%	86.0%	+17.0%	62.0%	75.0%	+13.0%
Jais-2-8B-Chat	66.0%	72.0%	+6.0%	66.0%	75.0%	+9.0%
ALLaM-7B	61.0%	69.0%	+8.0%	64.0%	78.0%	+14.0%

INSIGHT

Summary:

✔ Manual adjudication significantly improved the accuracy scores of all models.

✔ Notably, GPT-4o-mini achieved an outstanding 91.0% adjusted accuracy in zero-shot mode.

Key Takeaway: Many apparent "errors" by LLMs in Arabic NLP are actually **acceptable alternative formulations**, not true failures.

5.8 Limitations

While this benchmark provides valuable insights, several limitations must be acknowledged:

- **Dataset Size:** The evaluation was conducted on a relatively small, curated dataset of 100 sentences. While carefully balanced across dialects, a larger dataset would yield more statistically robust findings.
- **Subjectivity in Emotion:** Emotion detection is inherently subjective. Despite manual adjudication, some golden labels might be debated among different human annotators. Future iterations should involve multiple human annotators and calculate inter-annotator agreement (e.g., Cohen's Kappa).
- **Prompt Sensitivity:** LLMs are highly sensitive to prompt phrasing. The simple, standard prompts used (Zero-Shot and 3-Shot) might not elicit the maximum capability of every model, particularly for instruction-tuned models like Fanar that might require specific system prompt formats.

5.9 Final Conclusions and Recommendations

This benchmark evaluated five LLMs on Arabic emotion detection, a task that challenges models with linguistic diversity (MSA, Classical, Dialects) and nuanced emotion boundaries.

Final Conclusions:

1. **Proprietary vs. Open-Weights:** Proprietary models (**GPT-4o-mini** and **Gemini 2.5 Flash**) maintain a noticeable edge in zero-shot accuracy, particularly in handling dialectal nuances and adhering strictly to output formats. However, open-weights models like **ALLaM-7B** and **Jais-2-8B-Chat** demonstrate significant promise and competitive capability, especially when guided by few-shot examples.
2. **Impact of Few-Shot Prompting:** Interestingly, few-shot prompting did not universally improve performance across all models. While it significantly boosted ALLaM-7B (+7.0% raw accuracy), it actually caused a slight regression in GPT-4o-mini. This suggests that highly capable models might overfit to the specific few-shot examples, whereas smaller models use them to understand the task schema.
3. **The Necessity of Adjudication:** The manual adjudication process revealed that a large portion of apparent "misclassifications" by LLMs in Arabic are actually valid synonymous predictions (e.g., مفاجأة vs. اندهاش). Exact-match evaluation severely penalizes models for linguistic richness.

Recommendations:

- **Model Selection:** For zero-shot applications requiring high accuracy and out-of-the-box dialectal understanding, **GPT-4o-mini** (91.0% adjusted accuracy) is highly recommended. For environments where data privacy is paramount requiring an open-weights model, **ALLaM-7B** deployed with few-shot prompting is the most robust choice.
- **Evaluation Methodology:** Standard exact-match evaluation is fundamentally inadequate for generative LLMs. Future Arabic NLP benchmarks **must** incorporate semantic similarity metrics (using embeddings) or an "LLM-as-a-Judge" framework to automate the adjudication process fairly and account for synonymy.
- **Prompt Engineering for Constrained Outputs:** To mitigate the verbosity of open-weights models (which frequently output full sentences instead of the requested single-word label), future workflows should employ strict structural enforcement, such as JSON-schema generation constraints, to ensure parsing reliability.